Week 2 BA

2024 July07 Quiz-1

Comprehension for questions 1-5 (source Q5)

"Aqua Aqua (AA)" is a manufacturer of water purifiers. AA wants to understand the life of its RO filters. The life of a filter is measured in terms of "Litres of Water Purified (LWP)" before a warning light is raised for maintenance. A sample of 20 water purifiers were tested and the time (LWP) after which the warning light is raised for maintenance of the RO Filter was noted (data provided in Table-2). With this data, AA wants to understand the possible distribution for the life of the RO filter. To this end, it is proposed to bin the data in 4 groups: Low Life (LWP <= 25), Medium Life (25 < LWP <= 40), High Life (40 < LWP <= 80), and Excellent Life (LWP > 80). The following Φ values are given to you, where $\Phi(z)$ indicates the area under the curve (between "0" to "LWP") for a normal distribution with a mean of 60 and standard deviation of 24. $\Phi(-1.45) = 0.07$; $\Phi(-0.83) = 0.20$; $\Phi(+0.83) = 0.79$; $\Phi(+1.45) = 0.93$ Given this information, answer the subquestions

Sample-ID	RO Filter Life (in LWP)	Sample-ID	RO Filter Life (in LWP)
S-1	73	S-11	70
S-2	49	S-12	47
S-3	12	S-13	78
S-4	72	S-14	15
S-5	97	S-15	54
S-6	95	S-16	61
S-7	79	S-17	91
S-8	42	S-18	61
S-9	70	S-19	26
S-10	53	S-20	64

Table-2

1. [Source Q9; SA]
If the life of the RO filter is expected to follow a normal distribution with a mean of 60 and standard deviation of 24, then what is the expected frequency in the "Medium Life" bin? (Note-1: The bins DO NOT SPILT the sample space into equal areas) (Note-2: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 2.40 to 2.80]

2. [Source Q10; SA]
If the life of the RO filter is expected to follow a normal distribution with a mean of 60 and standard deviation of 24, then what is the expected frequency in the "Excellent Life" bin? (Note-1: The bins DO NOT SPILT the sample space into equal areas) (Note-2: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 4.10 to 4.30]

3. [Source Q11; SA]
If the life of the RO filter is expected to follow a normal distribution with a mean of 60 and standard deviation of 24, then what is the value of the computed test statistic? (Note-1: The bins DO NOT SPILT the sample space into equal areas) (Note-2: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 1.5 to 2.5]

4. [Source Q12; SA]
If the life of the RO filter is expected to follow a normal distribution with a mean of 60 and standard deviation of 24, then what is the degrees of freedom for the corresponding statistical test to verify the same? (Note-1: The bins DO NOT SPILT the sample space into equal areas) (Note-2: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

[Source ans: 3]

5. [Source Q13; MSQ]
Assume that a common value of 8.79 is identified for the tabulated test statistic (for both uniform and normal distribution tests), then (choose all that is applicable)

(a) REJECT the null hypothesis and conclude that the life of the RO filter is notUNIFORMLY distributed

(b) At the specified significance level, REJECT the null hypothesis and concludethat the life of the RO filter is not UNIFORMLY distributed

(c) REJECT the null hypothesis and conclude that the life of the RO filter is notNORMALY distributed

(d) At the specified significance level, REJECT the null hypothesis and concludethat the life of the RO filter is not NORMALY distributed

(e) DO NOT REJECT the null hypothesis and conclude that the life of the RO filteris UNIFORMLY distributed

(f) At the specified significance level, DO NOT REJECT the null hypothesis andconclude that the life of the RO filter is UNIFORMLY distributed

(g) DO NOT REJECT the null hypothesis and conclude that the life of the RO filteris NORMALY distributed

(h) At the specified significance level, DO NOT REJECT the null hypothesis andconclude that the life of the RO filter is NORMALLY distributed

[Source ans: b, h]

2024 Sep01 ET

6. [Source Q2; MSQ]
The computed p-value of a chi-square goodness-of-fit test is 0.067. Then, which of the following statements are TRUE

(a) At a 5% level of significance, the null hypothesis is rejected

(b) At a 5% level of significance, the null hypothesis is not-rejected

(c) At a 10% level of significance, the null hypothesis is rejected

(d) At a 10% level of significance, the null hypothesis is not-rejected

(e) At a 5% level of significance, the alternative hypothesis is rejected

(f) At a 5% level of significance, the alternative hypothesis is not-rejected

(g) At a 10% level of significance, the alternative hypothesis is rejected

(h) At a 10% level of significance, the alternative hypothesis is not-rejected

(i) None of these

[Source ans: b, d]

2024 Oct27 Quiz-1

7. [Source Q2; MSQ]

A hypothesis test is conducted to check if a data set follows a normal distribution. If the p-value is 0.67, then which of the following statements are **TRUE** (select all that is applicable)

(a) At a 5% level of significance REJECT the null hypothesis and conclude that thedata does not follow a normal distribution

(b) At a 5% level of significance DO NOT REJECT the null hypothesis and concludethat the data does not follow a normal distribution

(c) At a 5% level of significance REJECT the null hypothesis and conclude that thedata follows a normal distribution

(d) At a 5% level of significance DO NOT REJECT the null hypothesis and concludethat the data follows a normal distribution

(e) At a 10% level of significance REJECT the alternative hypothesis and concludethat the data does not follow a normal distribution

(f) At a 10% level of significance DO NOT REJECT the alternative hypothesis andconclude that the data does not follow a normal distribution

(g) At a 10% level of significance REJECT the alternative hypothesis and concludethat the data follows a normal distribution

(h) At a 10% level of significance DO NOT REJECT the alternative hypothesis andconclude that the data follows a normal distribution

[Source ans: d]

2024 Dec22 ET

Comprehension for questions 8-9 (source Q8)

The BS program coordinators wanted to understand the difficulty level of the BA course. Hence, past learners of the BA course were asked to give a rating (on a scale of 1 to 5, 1 meaning "least difficult" and 5 implying "most difficult"). A total of 200 students (100 male and 100 female) were approached for the survey and the result of the same is provided in Table-2. Given this data, answer the sub-questions.

Level of Difficulty for BA Course	Number of "Male Students" who opted for the level	Number of "Female Students" who opted for the level
1	15	30
2	10	10
3	20	15
4	40	20
5	15	25

Table-2

8. [Source Q12; SA]

Assuming that each difficulty level splits the probability space into equal areas, the general belief is that the level of difficulty across all learners follows a normal distribution. To validate this belief, what is the value of the computed test statistic? (Enter your answer in decimals rounded to two decimal places. Eg: If your answer is 10.256, then enter it as 10.26)

[Source ans: 21.00 to 22.00]

9. [Source Q13; SA]

For the hypothesis test focusing on whether the data is distributed normally, how many (count of) degrees of freedoms is present?

[Source ans: 2]

2025 Feb23 Quiz-1

10. [Source Q2; MCQ]
You are given a data of size 1000, which is centred at 50 and has a variance of 25. The median, mode, max and min values are 40, 22, 75 and 1 respectively.Then

(a) The data is right-tailed

(b) The data is left-tailed

(c) The data is symmetric

(d) None of these

[Source ans: a]

Comprehension for questions 11-14 (source Q10)

Say, Table 1 specifies the blood sugar level (in mg/dL) for the male and female patients who visit a hospital. It is assumed that the blood sugar level is distributed Normally with a mean of 50mg/ dL and standard deviation of 10 mg/dL. You are asked to perform your analysis by using the following bins. The bins are suggested such that the probability space is split into equal areas.

Bin-1: [0 mg/dL to 50 mg/dL]

Bin-2: [>50 mg/dL to 70 mg/dL]

Bin-3: [>70 mg/dL to 90 mg/dL]

Bin-4: [>90 mg/dL to 100mg/dL]

Patient ID	*Gender*	*Blood sugar level (in mg/dL)*
P1	MALE	71
P2	MALE	84
P3	MALE	76
P4	MALE	78
P5	MALE	89
P6	FEMALE	92
P7	FEMALE	97
P8	FEMALE	95
P9	FEMALE	83
P10	FEMALE	72

Then answer the given subquestions.

11. [Source Q11; SA]
What is the expected frequency in Bin-2 for the statistical test to be performed? (Note: give only a numerical value rounded to 2 decimal places. For example, if your answer is 10.1253, then input the answer as "10.13")

[Source ans: 2.5]

12. [Source Q12; SA]
What is the value of the computed test statistic for the statistical test to be performed? (Note: give only a numerical value rounded to 2 decimal places. For example, if your answer is 10.1253, then input the answer as "10.13")

[Source ans: 33]

13. [Source Q13; SA]
What is the number of degrees of freedom for the statistical test to be performed? (Note: give only a numerical value rounded to 2 decimal places. For example, if your answer is 10.1253, then input the answer as "10.13")

[Source ans: 3]

14. [Source Q14; MSQ]
If the p-value for the test is 0.85, then what can be the level of significance if the Null Hypothesis is NOT TO BE REJECTED (select all that is/are applicable)

(a) 0.9

(b) 0.15

(c) Any value between 0 and 1

(d) Cannot say without more data.

[Source ans: b]

2025 Apr13 ET

15. [Source Q2; MSQ]
Which of the following distributions is/are not symmetric in nature (choose all that are applicable)?

(a) Standard Normal distribution

(b) Standard Binomial distribution

(c) Uniform distribution

(d) Poisson distribution

[Source ans: d]

16. [Source Q3; MSQ]
Which of the following is/are required to build an empirical distribution? (choose all that are applicable)

(a) PDF or PMF

(b) Sample data

(c) Summary Statistics

(d) None of these

[Source ans: a, b, c]

2025 Jul13 Quiz-1

17. [Source Q4; MCQ]

For the summary statistics provided in Table-1, which of the following is true?

Parameter	Value
25th Quantile	3
50th Quantile	6
75th Quantile	8
First Moment	12
Second Moment	4
Third Moment	2

Table-1

(a) The distribution is right skewed

(b) The distribution is symmetric

(c) The distribution is left skewed

(d) Cannot say without the skewness value

[Source ans: a]

Comprehension for questions 18-23 (source Q6)

Milo's Wheels sells bicycles. DrMilo, the owner of the shop, is curious to know about his customer base. Hence, he collects data on the total sales made to different age groups in the year 2024. This data is given in Table-2. He hires you (a student in the BA course) as a data analyst and asks for your opinion. Then answer the given subquestions. (Note: For all questions in this comprehension, if your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")

Age group	Actual number of cycles sold
less than 10 years	25
11 to 15 years	50
16 to 20 years	30
21 to 24 years	50
25 to 30 years	20
31 to 40 years	25
greater than 41 years	50

Table-2

18. [Source Q7; SA]
Based on Table-2, you think the data is normally distributed such that the buckets split the distribution into areas that have equal probability. Then what is the probability of selling a cycle in any given bucket?

[Source ans: 0.13 to 0.15]

19. [Source Q8; SA]
Based on Table-2, you think the data is normally distributed such that the buckets split the distribution into areas that have equal probability. Then what is the expected number of cycles that will be sold to the age group of "25 to 30 years"?

[Source ans: 35 to 36]

20. [Source Q9; SA]
What is the value of the computed test statistic to perform a test to check if the Data in Table-2 follows a normal distribution?

[Source ans: 29 to 33]

21. [Source Q10; SA]
How many (count of) degrees of freedom is present for the performed test (to check if the data in Table-2 follows a normal distribution)?

[Source ans: 4]

22. [Source Q11; MCQ]
If the "Null Hypothesis" is rejected at a significance level of 0.1, then what will be the conclusion at a significance level of 0.5?

(a) Cannot say, insufficient data

(b) Null Hypothesis is rejected

(c) Null Hypothesis is not rejected

[Source ans: b]

23. [Source Q12; SA]
What is the sample size that DrMilo has provided for your analysis?

[Source ans: 250]

2025 Aug31 ET

Comprehension for questions 24-27 (source Q8)

The Following Comprehension and Related Subquestions are Purely Hypothetical A manufacturer produces lenses for sunglasses. In a given day, a total of 100 products were produced. Products produced usually can have scratches. The manufacturer is interested to see if the length of the scratches on produced lenses is normally distributed. Hence, based on the production of 100 lenses, he has identified the length of scratch on each lens and the grouped data is presented to you in Table-2. Further, from sample, the manufacturer has computed the standard deviation to be 1.00. Based on the production lot of 100 units (given sample), you are asked to check if the "Length of Scratches on produced lenses" follows a Normal Distribution. Given this information, answer the given subquestions.

Length of Scratches	Number of lenses with scratches of specified length
0.00 mm to 1.00 mm	30
1.01 mm to 2.00 mm	35
2.01 mm to 3.00 mm	15
3.01 mm to 4.00 mm	20

Table-2: Data on Number of products with scratches

24. [Source Q9; SA]
What would be the mean of the distribution, if you want to check the hypothesis "The length of scratches on a produced lens follows a Normal Distribution"? [Note: Enter your answer rounded to two decimal places. For example, if your answer is 1.235 then enter the answer as "1.24"]

[Source ans: 1.6 to 1.9]

25. [Source Q10; SA]
How many (count) of degrees of freedom will be presented for the computed test statistic for this problem if you want to check the hypothesis "The length of scratches on a produced lens follows a Normal Distribution"?

[Source ans: 1]

26. [Source Q11; SA]
What would be expected frequency in the last bin of the frequency table (Hint: The frequency table is obtained by only modifying Table-2 without adding any new bins)? [Note: Enter your answer rounded to two decimal places. For example, if your answer is 1.235 then enter the answer as "1.24"]

[Source ans: 18.00 to 19.00]

27. [Source Q12; MSQ]

Say the computed test statistic for the test is 12.3. Then, using Figure-2, at a significance level of 10% what will be concluded (choose all that are correct)?

ν \ α	.995	.990	.975	.950	.900	.500	.100	.050	.025	.010	.005
1	.00+	.00+	.00+	.00+	.02	.45	2.71	3.84	5.02	6.63	7.88
2	.01	.02	.05	.10	.21	1.39	4.61	5.99	7.38	9.21	10.60
3	.07	.11	.22	.35	.58	2.37	6.25	7.81	9.35	11.34	12.84
4	.21	.30	.48	.71	1.06	3.36	7.78	9.49	11.14	13.28	14.86
5	.41	.55	.83	1.15	1.61	4.35	9.24	11.07	12.83	15.09	16.75
6	.68	.87	1.24	1.64	2.20	5.35	10.65	12.59	14.45	16.81	18.55
7	.99	1.24	1.69	2.17	2.83	6.35	12.02	14.07	16.01	18.48	20.28
8	1.34	1.65	2.18	2.73	3.49	7.34	13.36	15.51	17.53	20.09	21.96
9	1.73	2.09	2.70	3.33	4.17	8.34	14.68	16.92	19.02	21.67	23.59
10	2.16	2.56	3.25	3.94	4.87	9.34	15.99	18.31	20.48	23.21	25.19

Figure-2: Chi-Square Table

(a) Reject the Null Hypothesis

(b) Do not Reject the Null Hypothesis

(c) Reject the Alternative Hypothesis

(d) Do not Reject the Alternative Hypothesis

(e) Accept the Null Hypothesis

(f) Accept the Alternative Hypothesis

[Source ans: a]

2025 Oct26 Quiz-1

28. [Source Q3; MSQ]
Which of the following distributions is/are Symmetric in nature (choose all that are applicable)?

(a) Standard Normal distribution

(b) Standard Binomial distribution

(c) Uniform distribution

(d) Poisson distribution

[Source ans: a, b, c]

2026 Mar15 Quiz-1

29. [Source Q1; MCQ]
A distribution is right-tailed if

(a) Coefficient of variation is positive

(b) Skewness is positive

(c) Skewness is negative

(d) Cannot say without the histogram

[Source ans: b]

2026 May06 ET

30. [Source Q48; MSQ]
Which of the following distributions is/are not symmetric distributions (select all that are applicable)?

(a) Standard normal distribution

(b) Exponential distribution

(c) Uniform distribution between [-1 to +1]

(d) Poisson distribution

[Source ans: b, d]